FX Variance Risk Premium in EUR/USD

A Vol-Selling Strategy with Macro Event and Regime Analysis

Motivation

Volatility measures the magnitude of price fluctuations in an asset over a given time period, independent of direction. Realized volatility (RV) captures historical price variation over the past 21 trading days, while implied volatility (IV) reflects the market's expectation of future fluctuations over the next 21-day window. RV is therefore a backward-looking measure whereas IV is forward-looking.

In practice, option buyers, corporates hedging FX exposure and portfolio managers purchasing downside protection, systematically overpay for insurance against adverse moves. This causes IV to consistently exceed subsequently realized RV, a phenomenon known as the Volatility Risk Premium (VRP). The seller of this insurance collects the premium as compensation for bearing tail risk: the strategy generates small, consistent profits in normal market conditions but is exposed to large losses in black swan events where RV dramatically exceeds IV.

EUR/USD is the world's most liquid FX pair, with daily volume exceeding $1 trillion. It is governed by two of the most systemically important central banks, the Federal Reserve and the European Central Bank, and its direction is heavily influenced by the interest rate differential between them. This makes EUR/USD an ideal market in which to study the VRP, and specifically to examine whether exposure to scheduled macro events such as FOMC and ECB meetings compresses the premium and drives tail risk.

Data & Methodology

Daily EUR/USD price data was sourced from Yahoo Finance spanning January 2018 to December 2024, providing six years of observations. Log returns were computed from daily closing prices, and a 21-day rolling standard deviation was calculated and annualised by multiplying by $\sqrt{252}$ to obtain realised volatility (RV). Implied volatility (IV) was extracted using the EVZ index ticker over the same date range and divided by 100 to express it in decimal form consistent with RV. The Volatility Risk Premium (VRP) was then defined as the daily difference $IV - RV$.

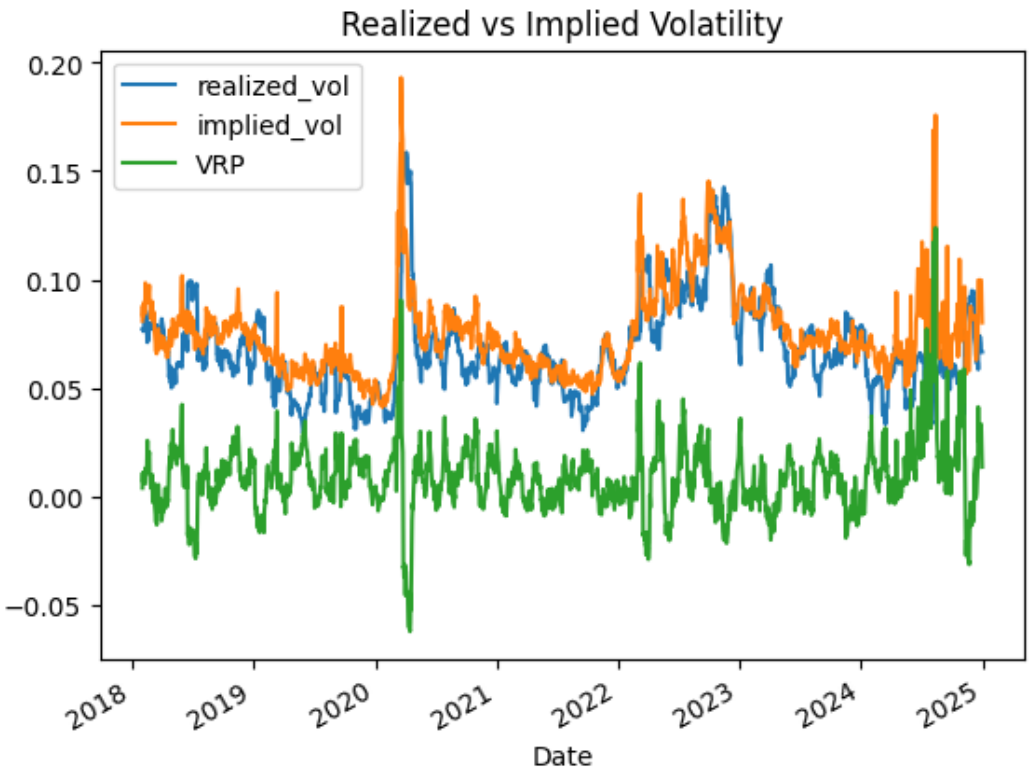


Figure 1: Realized vs Implied Volatility and VRP (EUR/USD, 2018-2024)

Several observations emerge from Figure 1. First, the spikes where RV dramatically exceeds IV correspond to black swan events, most visibly March 2020 at the onset of COVID-19. These represent the irreducible tail risk of a vol-selling strategy: events so sudden and severe that realised volatility outpaces even elevated implied volatility. No calendar-based filter can eliminate this risk. Second, a sharp positive VRP spike almost always precedes a sharp negative one. When a shock hits, IV reprices immediately as fear and uncertainty surge. RV follows with a lag, causing VRP to turn sharply negative. Paradoxically, the spike upward in VRP that precedes the crash can appear attractive to vol sellers, but as the data shows, the realised impact on volatility consistently exceeds the market's expectation in such events.

Base Strategy

The base strategy enters a vol-selling trade on any day where VRP exceeds its historical mean, on the premise that elevated IV relative to recent RV represents a mispricing the seller can exploit. P&L per trade is defined as IV at entry minus RV over the subsequent 21 trading days, reflecting the economics of selling a one-month straddle at current implied vol. This produced 820 trades over the sample period with a mean P&L of 0.0092 and an annualised Sharpe ratio of 1.50.

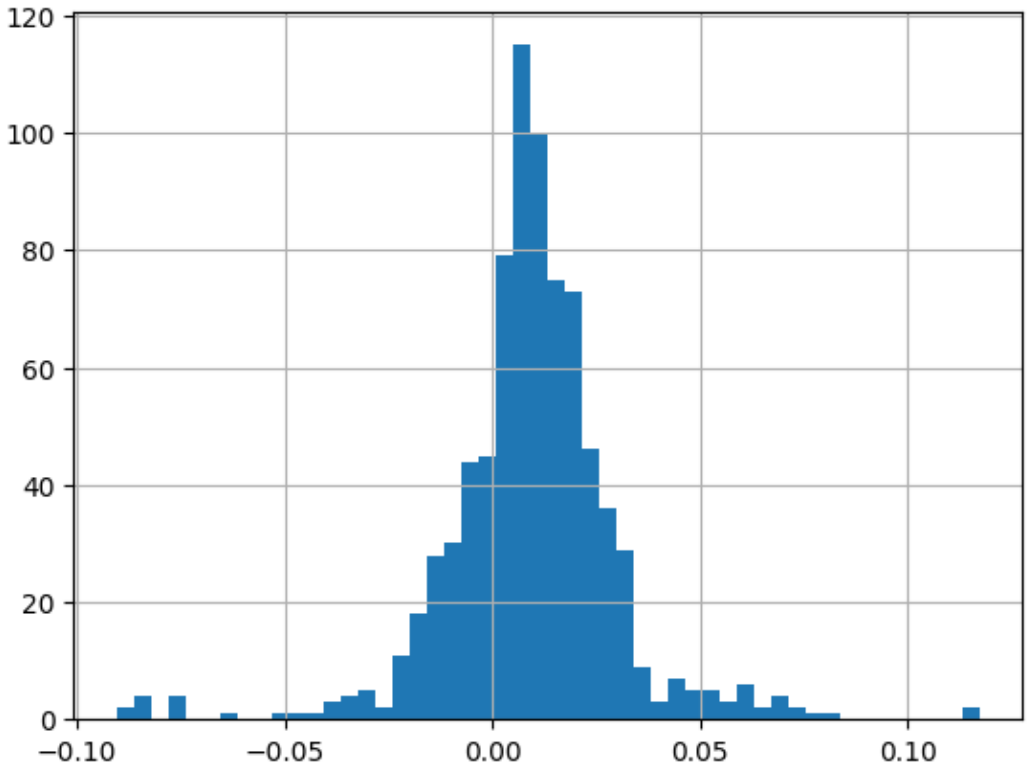


Figure 2: P&L Distribution - Base Strategy

The P&L histogram (Figure 2) confirms the characteristic payoff profile of a vol-selling strategy: a high frequency of small positive returns with a fat left tail of infrequent but large losses. The 5% VaR of -0.0198 indicates that in the worst 5% of trades, losses exceed 1.98 vol points. Given the distribution exhibits excess kurtosis of 8.24, well above the normal distribution value of 3, CVaR is a more appropriate risk measure as it captures expected loss severity beyond the threshold rather than merely the cutoff. The 5% CVaR of -0.0433 indicates that conditional on being in the worst 5% of outcomes, the expected loss is 4.33 vol points, more than double the VaR estimate, reflecting the fat-tailed nature of the distribution.

Fed Meeting Analysis

FOMC meeting dates were sourced publicly and each trade was tagged with a binary near_fed indicator, set to True if a Fed meeting fell within the 21-day holding window after entry.

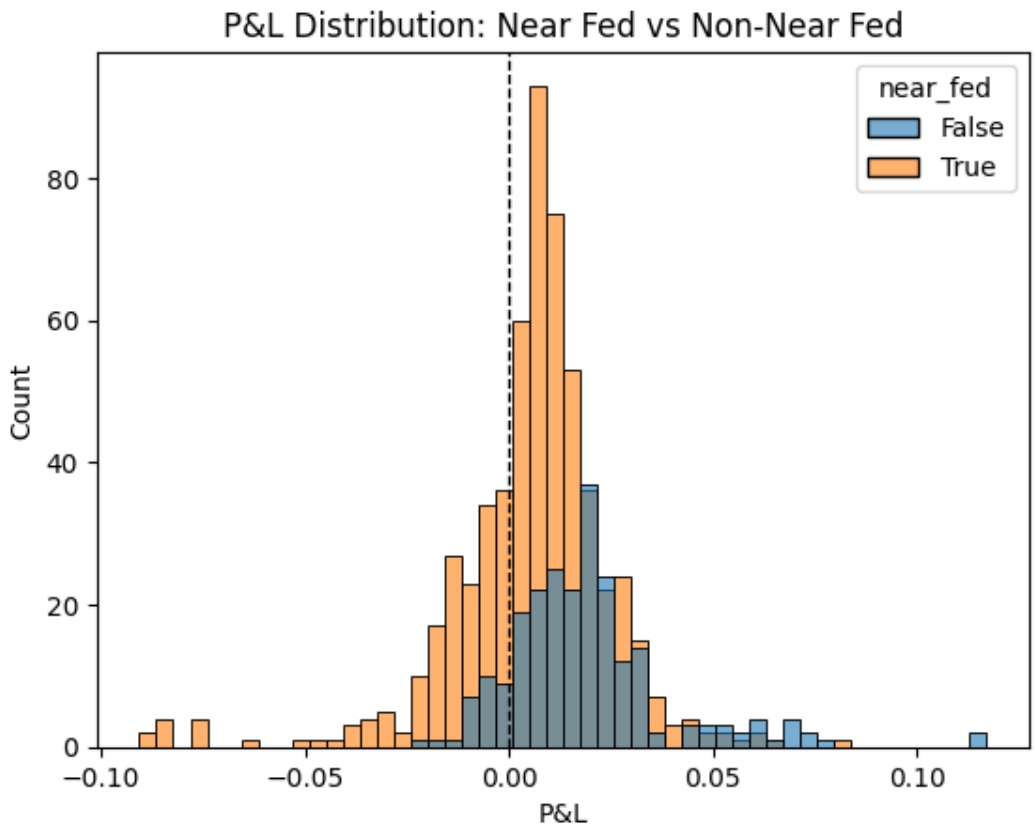


Figure 3: P&L Distribution by Fed Meeting Proximity

Figure 3 immediately reveals a clear pattern: virtually all large negative outliers belong to near-Fed trades, while the non-near-Fed distribution is concentrated in positive territory. The summary statistics confirm this:

Mean P&L (near Fed): 0.0053

Mean P&L (not near Fed): 0.0188

Percentage of trades near Fed: 70.98%

Mean P&L outside Fed windows is more than three times higher than during Fed windows, and notably higher than the base strategy mean of 0.0092, suggesting Fed proximity is a significant drag on performance. However, only 29.02% of trades fall outside Fed windows, raising two concerns. First, the sample size imbalance may affect statistical reliability. Second, since this is a premium-scraping strategy where trade frequency drives cumulative returns, restricting to non-Fed trades substantially reduces total P&L capacity.

To formally test whether the mean difference is statistically significant, a t-test is required. However, the standard t-test assumes independent observations, an assumption violated here because overlapping 21-day holding windows induce autocorrelation between consecutive trades.

A Durbin-Watson statistic of 0.311, well below the independence benchmark of 2, confirms this. To address this, we employ a Newey-West HAC regression with a lag of 21 days, treating near_fed as a binary regressor. This adjusts standard errors for both heteroscedasticity and autocorrelation, producing reliable inference. The results are as follows.

The near_fed coefficient of -0.0136 is significant at the 1% level ($t = -3.721$, $p < 0.001$), confirming that Fed meeting proximity reduces mean P&L by approximately 1.36 vol points per trade. Notably, the naive t-test produced a t-statistic of -8.69, more than double the Newey-West figure, illustrating how severely overlapping windows inflate apparent statistical certainty when autocorrelation is ignored.

ECB Meeting Analysis

The same analysis was repeated for ECB meeting dates, tagging each trade based on whether an ECB decision fell within its 21-day holding window.

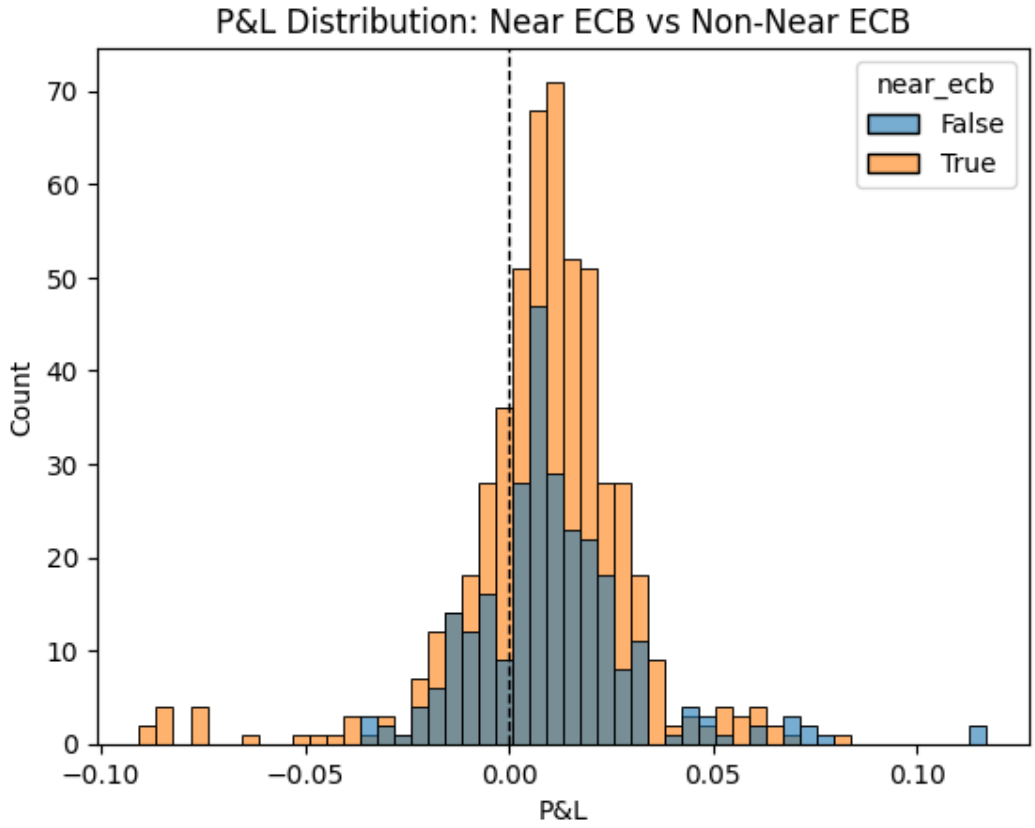


Figure 4: P&L Distribution by ECB Meeting Proximity

Figure 4 shows a less pronounced distributional difference between the two groups compared to the Fed analysis, with losses more evenly distributed across both near and non-near ECB trades. The summary statistics reflect this:

Mean P&L (near ECB): 0.0084

Mean P&L (not near ECB): 0.0107

Percentage of trades near ECB: 65.61%

The mean difference of 0.0023 is substantially smaller than the Fed equivalent of 0.0135, suggesting ECB proximity has a more limited impact on strategy performance. Applying Newey-West HAC regression with a 21-day lag confirms this: the near_ecb coefficient of -0.0023 fails to reach significance at either the 5% or 1% level ($t = -0.639$, $p = 0.523$). We therefore cannot reject the null hypothesis that ECB meeting proximity has no effect on mean P&L.

This is quite a meaningful finding. EUR/USD is priced in USD, and the Federal Reserve's decisions affect global dollar liquidity (the world's reserve currency), creating systemic repricing across all dollar-denominated assets. ECB decisions primarily affect the Eurozone. While the interest rate differential between the two central banks is a key driver of EUR/USD direction, it is Fed decisions that disproportionately drive volatility spikes in this pair. Consequently, filtering on ECB dates is not statistically justified and would unnecessarily reduce trade frequency without a commensurate improvement in risk-adjusted returns.

Volatility Regime Analysis

Trades were classified into high and low volatility regimes using the median implied volatility across all 820 base trading days as the threshold. The median was chosen over the mean to avoid distortion from extreme outliers such as the COVID spike in March 2020. This produced an even split by construction, with 406 high-vol and 414 low-vol trades.

The mean P&L difference between regimes is substantial:

Mean P&L (high vol regime): 0.0136

Mean P&L (low vol regime): 0.0049

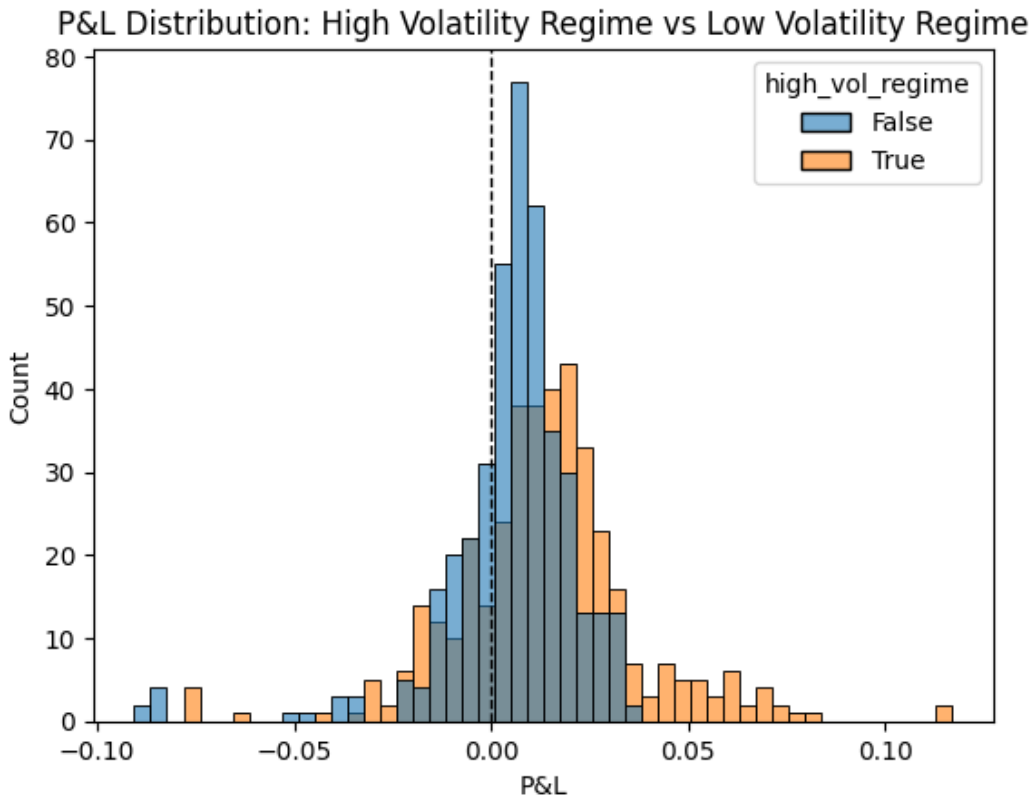


Figure 5: P&L Distribution by Volatility Regime

Figure 5 reinforces this: the right tail of the distribution is dominated by high-vol regime trades, while the low-vol distribution is tightly clustered near zero with minimal upside. Applying Newey-West HAC regression with a 21-day lag confirms statistical significance at the 5% level (coefficient = 0.0087, t = 2.382, p = 0.017), though notably not at the 1% level, reflecting greater uncertainty than the Fed finding.

The economic intuition is that the VRP is countercyclical. In high IV regimes, markets are fearful and options buyers are willing to overpay for protection, and the premium collected by the vol seller is larger. In most cases, this elevated fear is not fully realised: RV does not reach the level IV implied, and the seller profits. However, Figure 5 also shows that some of the worst losses also occur in high-vol regimes. When a shock is severe enough that RV dramatically exceeds even elevated IV (as in March 2020), the high-vol regime that appeared attractive becomes the source of maximum loss. This is the fundamental asymmetry of the strategy: high-vol regimes offer the largest edge while also concentrating the tail risk.

Annual Performance Breakdown

Decomposing base strategy performance by year reveals meaningful variation driven by the macro environment.

2020 and 2022 produced the lowest Sharpe ratios at 0.09 and 0.27 respectively. The COVID shock in March 2020 caused a sudden and severe RV spike that outpaced even elevated IV,

generating the strategy's worst losses. Notably, 2020 also had the highest trade count (141), reflecting that IV was frequently above its mean as markets priced in persistent uncertainty, a double-edged dynamic where high trade frequency coincided with elevated tail risk. In 2022, the Fed's aggressive rate hiking cycle, the fastest tightening since the 1980s, created sustained, persistent realised volatility rather than a single recoverable spike. With RV elevated for months, the IV-RV spread compressed consistently throughout the year, eroding the strategy's edge.

2019 and 2024 were the strongest years with Sharpe ratios of 1.11 and 0.82. Both represent periods of relative macro stability where IV remained elevated due to institutional demand for hedging while RV normalised, precisely the environment a vol seller seeks. 2024 is particularly notable as it combined the highest mean P&L (0.0208) with the highest trade count (160), suggesting the post-hiking stability created both a wider premium and more frequent trading opportunities.

The strategy remained profitable in every year of the sample, even in its worst years, a meaningful result given the tail risk inherent to vol selling.

Strategy Comparison

Four strategies are compared to assess the impact of the filters identified in the preceding analysis:

- Base: all trades where VRP exceeds its mean
- Exclude Fed: base strategy excluding trades with a Fed meeting in the holding window
- High Vol Regime: base strategy restricted to high IV regime trades only
- Exclude Fed & High Vol: combining both filters

ECB dates were excluded from filtering given the statistically insignificant results from the ECB analysis.

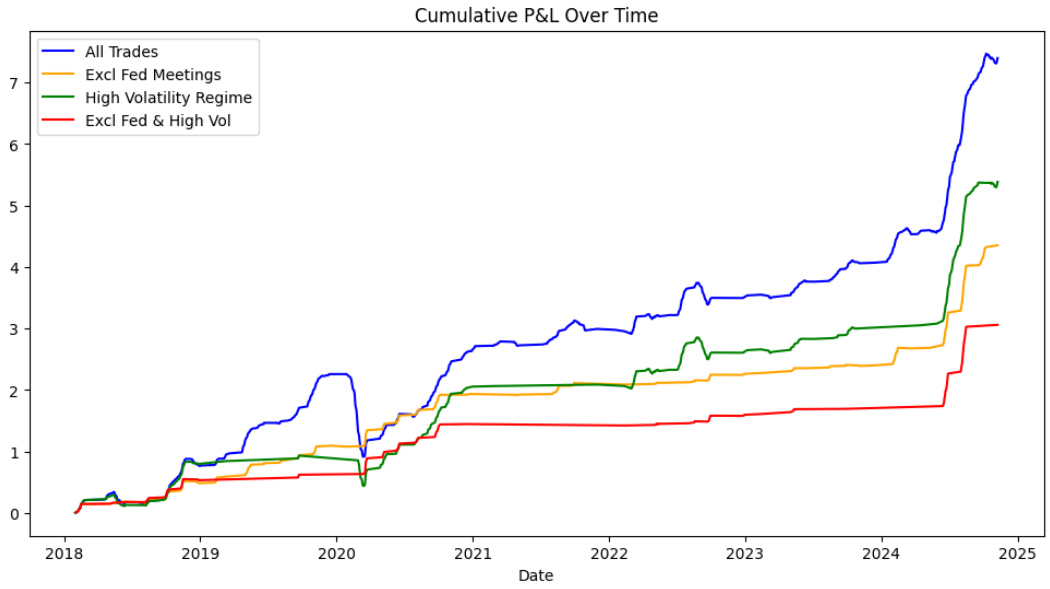


Figure 6: Cumulative P&L Over Time by Strategy

Figure 6 plots cumulative P&L over time for all four strategies. The base strategy produces the highest total cumulative P&L by end of sample, driven purely by trade volume. The 2020 drawdown is clearly visible in both the base and high-vol strategies. The sharp drop in early 2020 represents the COVID shock, with strategies incorporating the Fed exclusion filter showing considerably more resilience through this period.

Strategy	**Trades**	**Mean P&L**	**Annualised Sharpe**	**5% CVaR**
Base	820	0.0092	1.50	-0.0433
Excl Fed	238	0.0188	3.28	-0.0111
High Vol	406	0.0136	1.98	-0.0395
Excl Fed & High Vol	132	0.0245	3.64	-0.0084

Cumulative P&L alone is misleading here: the base strategy's higher total reflects trade volume rather than edge quality. On a risk-adjusted basis, Strategy 4 dominates across both mean P&L and Sharpe, improving mean P&L by 167% and Sharpe by 142% relative to the base. The tradeoff is a reduction from 820 to 132 trades over six years, roughly 22 trades per year, which raises practical concerns around strategy capacity and transaction costs. In live trading, bid-ask spreads on EUR/USD straddles would materially erode per-trade P&L, making trade frequency a critical consideration.

The CVaR column is particularly revealing: excluding Fed meetings alone reduces expected tail loss by 74% (from -0.0433 to -0.0111), while the combined filter reduces it by 81% to -0.0084. This demonstrates that the Fed filter is not merely improving average returns but is directly eliminating the trades responsible for the worst losses in the distribution. The high-vol only filter improves mean P&L and Sharpe but leaves CVaR largely intact at -0.0395, confirming that tail risk in this strategy is driven primarily by macro event exposure rather than vol regime alone.

Limitations & Extensions

Several limitations of this analysis should be acknowledged. First, the mean VRP threshold used in the base strategy entry condition and the median IV threshold used for regime classification are both computed over the full sample period. In live trading, these parameters would not be available in advance. A robust implementation would instead use rolling or expanding window estimates calibrated solely on past data. Testing strategy performance under this constraint is an important avenue for future work.

Second, transaction costs are excluded from the model. Bid-ask spreads on EUR/USD straddles can be 0.2-0.5 vol points, which would materially erode a mean P&L of 0.009 per trade. That said, Strategy 4 is likely most resilient to this concern: its combination of the highest mean P&L and lowest trade count minimises total commission costs relative to returns.

Third, the analysis is limited to a single instrument. It is unclear whether the findings generalise across currency pairs or whether the Fed meeting effect is specific to USD pairs. Extending the analysis to USD/JPY would be particularly interesting given the Japanese yen's significance in global debt markets and its sensitivity to US rate differentials.

On the ECB finding, the 30-day forward window may be too broad to detect a meaningful effect. A refined specification targeting trades with an ECB meeting in the final days of the holding period, when the impact on realised vol is most direct, may yield stronger results.

Finally, the current strategy uses binary entry signals. A natural extension would be to scale position sizes dynamically by the magnitude of VRP at entry, increasing exposure when the premium is largest and reducing it in marginal cases. This would improve capital efficiency without changing the fundamental strategy logic.